

Michael McLain

AI & Software Engineer

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PROFILE

Software and DevOps engineer who turns manufacturing data into tools that quote, predict and trace, and runs the infrastructure they live on. Nearly four years at a Tier 1 supplier of carbon fibre structures and interiors for ultra-premium OEM programmes (McLaren W1, Aston Martin Valhalla and Valkyrie, Mercedes-AMG ONE, Gordon Murray T.33). Shipped 30+ internal tools, cut quoting from 4 hours to 30 minutes per car, took carbon nesting efficiency from 40% to 90%, and own a five-site estate end to end. Also take on projects through McLain & Associates.

EXPERIENCE

TECNIQ Ltd, Essex Dec 2022 – present · 3 yrs 10 mos

DevOps & Software Engineer Dec 2023 – present · 2 yrs 10 mos

- Built QuoteDesk, the quoting engine for a bespoke-options OEM programme: spec-pack import, rules-based pricing with certainty-scored proposals, branded PDF output. Quote time per car went from 4 hours to 30 minutes.
- Built STEP Quoter: extracts geometry features from a dropped CAD file, scores complexity and prices against a dataset of ten costed programmes (roughly 8,000 parts with geometry), with margin visible before anything is sent. 30 to 40 people across commercial and engineering use the two tools daily.
- Replaced a manual Excel and VBA timesheet approval process with a web app: Microsoft sign-in, spreadsheet ingest, per-director exports for 21 approvers.
- Led the MRP rollout across two OEM programmes, then built the analytics on top of it: daily production breakdowns, departmental target-vs-actual dashboards and executive reports from raw MRP exports. Now leading the PLM migration from SmarTeam to Dassault 3DEXPERIENCE.
- Own the estate across 5 sites: rebuilt 13 VMs from bare metal, redesigned a flat network into a segmented multi-site one, refreshed 100+ devices including custom-built CAD workstations, and run Microsoft 365, Entra ID, CrowdStrike, Cyber Essentials and TISAX.
- Scoped and costed on-prem LLM inference on Apple Silicon so OEM data can be used with AI without leaving the building.

Software & Data Engineer Dec 2022 – Nov 2023 · 1 yr

- Carbon kit-cutting waste project: analysed cut data across the programmes, specified a new kit-cutting machine and introduced AI-driven nesting. Material efficiency went from 40% to 90%.
- Introduced digital traceability on the shop floor with 2D Data Matrix marking, so every part carries its history from ply to assembly.
- Took on the IT estate alongside development and laid the groundwork for the network and virtualisation rebuild.

SELECTED PROJECTS

QuoteDesk Live · Next.js 14, TypeScript, Supabase, PDF generation
Quoting automation for bespoke OEM interior options: spec-pack import, part matching that learns renumbered parts, per-part pricing rules with certainty scoring, branded PDF output and a full audit trail.

STEP Quoter In beta · Next.js, TypeScript, Supabase, geometry analysis
Drop a STEP file in, get a costed quote: geometry-derived complexity, historic costings plus labour, machine, material and supplier rates, and cost, sale price and margin side by side before a quote goes out.

DatumQMS In development · Next.js 15, React 19, Tailwind 4
Own product: a lightweight IATF 16949 / PPAP compliance app for small automotive suppliers. PPAP submissions as structured data, not documents. Working demo built; pilot in discussion.

SKILLS

Software	TypeScript, React, Next.js, Tailwind, Node, Supabase, Postgres, SQL Server, C# / .NET, Go
Data & AI	Python, pandas, SQL, pricing models, geometry feature extraction, LLM integration
Infrastructure & ops	Vercel, Hyper-V, XCP-ng, UniFi, Microsoft 365, Entra ID, CrowdStrike, Cyber Essentials, TISAX
Manufacturing	MRP (Progress Plus), PLM (SmarTeam, 3DEXPERIENCE), IATF 16949 / PPAP, Data Matrix traceability, AI nesting

EDUCATION

BEng Software Engineering, First Class Honours Sep 2019 – Jun 2022
Canterbury Christ Church University, Canterbury